DAILIN CUN

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EDUCATION BACKGROUND

Beijing University of Chemical Technology (BUCT) **Beijing, China**
B.Eng in Electronic Science and Technology Sept. 2022 - Jun. 2026
Main courses: *Digital Electronic Technique, Analog Electronic Technique, Signals and Systems, Optoelectronic Technology, Microwave Technique, Quantum Mechanics, Solid-State Physics, Semiconductor Physics*

Chinese University of Hong Kong (CUHK) **HongKong, China**
Msc in Microelectronics and Integrated Circuits Sept. 2026 - Nov. 2027
Main courses: *Adv.CMOS Analog IC Design, DSP IC Design, Digital IC Deign & AI IC, RISC-V Integrated Circuit, AI IC Design, Smart Power IC Deign, Adv.Semiconductor Packaging,*

PROJECT EXPERIENCE

Portable Heart Rate Monitoring System Based on STC89C52 Microcontroller, BUCT lab **Beijing, China**
Research Assitant May. 2025 - Jul. 2025

- Built a hear rate monitoring system using ST188 reflective optical sensor, STC89C52 MCU, and LCD1602 display;
- Designed analog signal conditioning circuitry including pre-amplification and active bandpass filtering to exact pulses for raw sensor output.
- Developed firmware in C using Keil, implementing real-time heart rate calculation from inter-beat intervals and buzzer alert generation for abnormal rates.
- Designed PCB, hand-soldered the full prototype, and validated through simulation and physical testing;

NPN Transistor Output Characteristic Curve Measurement System, BUCT lab **Beijing, China**
Team Leader Aug. 2024 - Sept. 2024

- Designed and simulated an automatic NPN transistor output characteristic curve measurement system in Multisim, integrating LM555, LM324, 74161, and logic circuits.
- Implemented automatic collector-voltage sweeping and stepped base-current biasing to obtain NPN transistor output characteristic curves.
- Designed differential amplifier circuitry for collector-current sensing and X–Y oscilloscope visualization.
- Performed circuit analysis, waveform verification, and parameter tuning in Multisim to validate system functionality.

WORK EXPERIENCE

Dehong Radio and Television Safety Broadcasting Monitoring Center **Yunnan, China**
Broadcast Signal Monitoring Intern Jul. 2024 - Aug. 2024

- Monitored live broadcast signals to ensure stable transmission across multiple radio and television channels.
- Logged transmission issues, including video stuttering, mosaic artifacts, and audio interruptions, and prepared monitoring reports.
- Assisted with field signal testing using spectrum analyzers to evaluate signal strength and reception quality.

COMPETITIONS EXPERIENCE

"Siemens Cup" China Intelligent Manufacturing Challenge (CIMC) **Beijing, China**
Team Leader Aug. 2024 - Sept. 2024

- Designed an industrial Ethernet network architecture for an intelligent manufacturing production line, including VLAN planning, hierarchical IP address allocation, firewall configuration, PLC communication debugging, and network performance analysis using Wireshark.
- Won **Second Prize (North China Division)** in the Innovative Design of Industrial Network Structures and Functions competition.

MORE AWARDS

- **Honorable Mention,** Mathematical Contest in Modeling (MCM/ICM), 2025
- **First Prize (Beijing Division),** China Undergraduate Mathematical Contest in Modeling (CUMCM), 2024
- **Third Prize (National),** Asia and Pacific Mathematical Contest in Modeling (APMCM), 2024
- **Quality Development Competition Scholarship**, 2024
- Government Scholarship (**Third Class**), 2023-2024 Academic Year

ADDITIONAL INFORMATION

LAUNGUAGE: Chineses (Native), English (IELTS 6.5)
Programming & Embedded Systems: Verilog, C, Python
Hardware & Circuit Design: Altium Designer, Multisim, Proteus, JLCPCB prototyping, Quartus, Keil, MATLAB
Design & Media Tools: Photoshop, Blender (3D modeling), Xiumi (social media design)